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| **Fiscal Year:** | FY 2024 | **Task Last Updated:** | FY 12/15/2023 |
| **PI Name:** | Carter, Dorothy Ph.D. | | |
| **Project Title:** | Project FUSION: Facilitating Unified Systems of Interdependent Organizational Networks--Grant 80NSSC23K1124 | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 48824 | **Congressional District:** | 7 |
| **Comments:** | The Principal Investigator (PI) was previously at the University of Georgia. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship |
| **Start Date:** | 02/12/2018 | **End Date:** | 08/21/2026 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 4 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
| **Contact Email:** | alexandra.m.whitmire@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | Key Personnel Changes: added Alina Lungeanu, added Laura Bauer, added Caton Weinberg, added Alek Roehl. These personnel changes occurred on October 12, 2023. [Ed. Note: Drs. Lungeanu, Bauer, Weinberg, and Roehl were added as Consultants to this project.] | | |
| **COI Name (Institution):** | Shuffler, Marissa L ( Clemson University )<br>DeChurch, Leslie ( Northwestern University )<br>Contractor, Noshir ( Northwestern University )<br>Schecter, Aaron ( University of Georgia ) | | |
| **Grant/Contract No.:** | 80NSSC23K1124 ; 80NSSC18K0511 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | As we set our sights on Mars, and other destinations beyond lower Earth orbit, we must enable extreme forms of teamwork across Spaceflight Multiteam Systems (SFMTSs) composed of teams that are separated by unprecedented degrees of space and time. In “Project FUSION: Facilitating Unified Systems of Interdependent Organizational Networks,” we are engaging in a transformative research program rooted in the past decade of theory and research on MTSs, but breaking new ground in how MTSs are conceptualized and studied. Our programmatic research aims to illuminate the underlying forces that give rise to the psycho-social relational states (e.g., influence, trust, shared cognition) within and between teams that underpin mission success. These crucial relationships, and the drivers of their emergence, will need to be understood, monitored, and at times, circumvented using countermeasures in order to enable coordinated efforts across the SFMTSs involved in Long-duration Exploration Mission (LDEMs). Project FUSION is a multi-pronged, multi-method, interdisciplinary project with three main research foci: (1) field investigations using NASA personnel; (2) development of an agent-based computational model capturing the drivers of relational states; and (3) controlled laboratory experiments and analog studies. Our research design is intended to be iterative. Findings within each foci are continually infusing the refinement and design of research in other foci. Further, Project FUSION is an applied research project with the ultimate goal of translating findings from three research foci in order to provide NASA with a "countermeasure toolkit" comprised of validated interventions that can be used to facilitate effective teamwork in SFMTSs. The countermeasure toolkit under development in this project consists of: (1) a SFMTS task analysis procedure, (2) a decision-making guidebook based on our agent-based computational model of SFMTS dynamics, (3) a multiteam training countermeasure ready for operational implementation with astronauts and mission controllers, and (4) a multiteam debriefing protocol ready for operational implementation with astronauts and mission controllers. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The findings from this project will have substantial implications for human life on Earth, and in particular, for the effectiveness of teams and larger systems of teams in organizations operating in high-stakes environments. The field studies, laboratory studies, and computer simulation studies aim to better understand the patterns of social relationships (e.g., trust, influence, information sharing) that are likely to form within and across teams in large interdependent organizational systems. By better understanding the patterns of relationships that are likely, we can help determine when and where teamwork interventions or "countermeasures" are necessary. Moreover, the ultimate goal of this applied research project is to develop and validate a toolkit of countermeasures, including training, debriefing procedures, and decision-making protocols that are designed to facilitate team and inter-team collaboration in complex organizational systems. These countermeasures will be able to be utilized widely across many organizational contexts beyond NASA (e.g., healthcare, the military, corporations). |
| **Task Progress:** | NOTE: Continuation of Project FUSION, Grant #80NSSC18K0511 (PI: Carter). See previous grant for earlier report information (Ed., 1/24/24).<br>As we set our sights on Mars we must enable synchronized human efforts across complex ‘multiteam systems’ (MTSs) composed of numerous teams that are separated by unprecedented degrees of space and time. Project FUSION is a programmatic stream of research that aims to identify the key factors affecting networks of relationships within and between teams in Spaceflight MTSs (Aim #1), the key developmental factors that could trigger shifts in these relational networks (Aim #2), the impact relational networks have on Spaceflight MTS system coordination and performance (Aim #3), and deliver a set of validated proactive and/or reactive countermeasures targeting relational networks in order to support Spaceflight MTS coordination and mission success in long-duration exploration missions (LDEM) (Aim #4).<br><br>To achieve these aims, we are connecting findings from three Research Foci. Research Foci #1 includes reviews of relevant academic literature, analyses of archival documents, and interviews and observations with NASA personnel and personnel in analog environments (e.g., hospital systems). In Research Foci #2, we are developing agent-based computational models of spaceflight MTS functioning during LDEM and using these models to conduct ‘virtual experiments’ to understand how to best support MTS functioning. In Research Foci #3, we are conducting a series of human-subjects experiments in university laboratories, space analog environments, and classroom settings.<br><br>In the past year, we advanced Research Foci #1 in multiple ways, including: (1) conducting a series of interviews with former members of NASA’s Mission Control Center (MCC) to understand how specific roles in the spaceflight MTS may need to shift to meet the needs of future missions; (2) analyzing critical incidents related to MTS functioning, drawn from NASA archival documents to understand the coordination mechanisms spaceflight MTSs use to adapt to complex challenges; and (3) published an article about conducting effective pre/de-brief MTS meetings in an analogous high-reliability context (a hospital system). We advanced Research Foci #2 by leveraging the data collected in Foci #3 to develop a preliminary computational model that predicts how members of spaceflight MTSs problem-solve under conditions of communication delay. We advanced Research Foci #3 by completing human-subjects data collection within the NASA Human Exploration Research Analog (HERA) Campaign 6 (C6). Our analyses of HERA C5 and C6 data indicate that communication delays significantly disrupt problem-solving networks among members of Spaceflight MTSs and these disruptions negatively impact MTS performance. |
| **Bibliography Type:** | Description: (Last Updated: 08/05/2025) |
| **Books/Book Chapters** | Pearman J, Gerkin E, Carter DR. "What is ‘functional’ about distributed leadership in teams?" in "The SAGE Handbook of Leadership." Ed. D. Schedlitzki, M. Larsson, B. Carroll, M.C.Bligh, O. Epitropaki. London: SAGE Publications Ltd, 2023. pp. 72-82. https://doi.org/10.4135/9781529781892 , Mar-2023 |